

# SIMATS ENGINEERING

## ASSESSMENT TOOL 1

**COURSE NAME: OBJECT ORIENTED ANALYSIS AND DESIGN**  
**COURSE CODE: CSA11**

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### **COURSE OUTCOME COVERED**

**CO1:** *Apply object-oriented concepts and software development life cycle models to design structured and modular software systems. (BL2, BL3)*

*Assessment Tool Weightage: 60%*

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Dr Ramamoorthy

**QUESTIONS: 12**

**Total Marks:60**

Annexure A

### **CONCEPT MAPPING**

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#### ***Code of Conduct:***

*I E. SAKTHI PRIYA ( 192520041 ) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

#### ***Signature of the student:***

E. SAKTHI PRIYA

Annexure A

**CONCEPT MAPPING**

## 1. Construct a Concept Map

**Given:**

Requirement Analysis → \_\_\_\_\_ → Coding → Testing → \_\_\_\_\_ → Maintenance

**Answer:**

The missing concepts are:

- Design
- Deployment

**Explanation:**

- Design converts requirements into system architecture.
- Deployment releases the software to end users.

**Analysis:**

- Ensures systematic software development.
- Reduces project risks.
- Improves software quality and maintainability.

**(5 Marks)**

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## 2. Construct a Concept Map

**Given:**

Object-Oriented Principles

→ Abstraction

→ \_\_\_\_\_

→ Inheritance

→ \_\_\_\_\_

**Answer:**

The missing concepts are:

- Encapsulation
- Polymorphism

**Explanation:**

- Encapsulation protects data from unauthorized access.
- Polymorphism allows one interface to support multiple behaviors.

**Analysis:**

- Improves security and flexibility.
- Enhances code reusability.
- Supports efficient software design.

**(5 Marks)**

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### 3. Construct a Concept Map

**Given:**

Class → Blueprint

Object → Instance

Method → \_\_\_\_\_

Attribute → \_\_\_\_\_

**Answer:**

The missing concepts are:

- Behavior
- State

**Explanation:**

- Methods define the behavior of objects.
- Attributes represent the state of objects.

**Analysis:**

- Helps model real-world entities.
- Defines object characteristics and actions.
- Improves system organization.

**(5 Marks)**

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## **4. A Concept Map Shows**

**Given:**

Software Quality

→ Reliability

→ \_\_\_\_\_

→ Maintainability

**Answer:**

The missing concept is:

- Usability

**Explanation:**

- Usability measures how easily users interact with software.
- Reliable software performs consistently.
- Maintainable software is easy to update.

**Analysis:**

- Improves user satisfaction.
- Enhances product acceptance.
- Contributes to software success.

**(5 Marks)**

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## **5. UML Concept Map**

**Given:**

UML

→ Structural Diagrams → \_\_\_\_\_

→ Behavioral Diagrams → Activity Diagram

→ Interaction Diagrams → \_\_\_\_\_

**Answer:**

The missing concepts are:

- Object Diagram
- Communication Diagram

**Explanation:**

- Object diagrams represent object instances and relationships.
- Communication diagrams show message exchange among objects.

**Analysis:**

- Improves system visualization.
- Supports object-oriented analysis.
- Facilitates stakeholder communication.

**(5 Marks)**

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**6. Given:**

Requirements ✓ → Design ✓ → Coding ✗ → Testing ✗ → Deployment ✗

**Explanation:**

- Failure occurs during the Coding phase.
- Program logic contains implementation errors.

**Analysis:**

- Errors affect subsequent phases.
- Testing cannot proceed effectively.
- Software quality decreases.

**Solution:**

- Review source code.
- Apply coding standards.
- Conduct code inspections before testing.

**(5 Marks)**

## 7. Construct a Concept Map

**Given:**

Problem Identification → \_\_\_\_\_ → Design → Implementation → \_\_\_\_\_

**Answer:**

The missing concepts are:

- Requirement Analysis
- Testing

**Explanation:**

- Requirement Analysis identifies user needs and system requirements.
- Testing verifies whether the implemented system meets the specified requirements.

**Analysis:**

- Improves software quality.
- Reduces development errors.
- Ensures successful project completion.

**(5 Marks)**

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## 8. Construct a Concept Map

**Given:**

Object-Oriented Features

→ Class

→ \_\_\_\_\_

→ Message Passing

→ \_\_\_\_\_

**Answer:**

The missing concepts are:

- Object
- Dynamic Binding

**Explanation:**

- Objects are instances of classes.
- Dynamic binding allows method calls to be resolved at runtime.

**Analysis:**

- Supports flexibility in software design.
- Improves code maintainability.
- Enables runtime polymorphism.

**(5 Marks)**

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## 9. Construct a Concept Map

**Given:**

Software Development Models

→ Waterfall

→ \_\_\_\_\_

→ Spiral

→ \_\_\_\_\_

**Answer:**

The missing concepts are:

- Incremental Model
- Agile Model

**Explanation:**

- Incremental development delivers software in stages.
- Agile supports iterative development with continuous customer feedback.

**Analysis:**

- Enhances adaptability.
- Reduces project risks.
- Improves customer satisfaction.

**(5 Marks)**

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## 10. Construct a Concept Map

**Given:**

Class Diagram

→ Class

→ \_\_\_\_\_

→ Association

→ \_\_\_\_\_

**Answer:**

The missing concepts are:

- Attribute
- Method

**Explanation:**

- Attributes define the properties of a class.
- Methods define the operations performed by a class.

**Analysis:**

- Represents object structure effectively.
- Improves software documentation.
- Supports object-oriented design.

**(5 Marks)**

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## 11. Construct a Concept Map

**Given:**

Software Engineering Principles

→ Modularity

→ \_\_\_\_\_

→ Reusability

→ \_\_\_\_\_

**Answer:**

The missing concepts are:

- Maintainability
- Scalability

**Explanation:**

- Maintainability simplifies software updates.
- Scalability enables systems to handle increased workload efficiently.

**Analysis:**

- Improves software performance.
- Reduces maintenance effort.



- Supports long-term software evolution.

**(5 Marks)**

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## **12. Given:**

Requirement Analysis ✓ → System Design ✓ → Coding ✓ → Integration ✗ →  
Deployment ✗

### **Explanation:**

- Failure occurs during the Integration phase.
- Individual modules fail to communicate correctly.

### **Analysis:**

- Causes functional inconsistencies.
- Delays software deployment.
- Increases debugging effort.

### **Solution:**

- Perform integration testing.
- Verify module interfaces.
- Resolve compatibility issues before deployment.

**(5 Marks)**

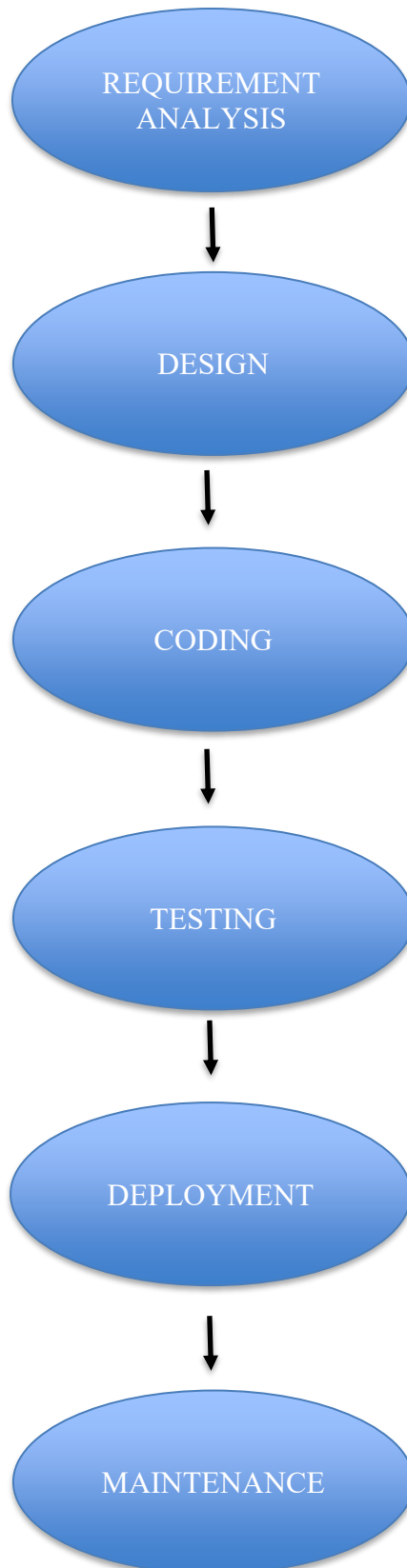
## CONCEPT MAPPING

### RUBRICS FOR CALCULATION

#### CONCEPT MAPPING

| Criteria                  | Weightage | Level 4 (Exemplary)                                                         | Level 3 (Proficient)                              | Level 2 (Developing)                                        | Level 1 (Beginning)                     |
|---------------------------|-----------|-----------------------------------------------------------------------------|---------------------------------------------------|-------------------------------------------------------------|-----------------------------------------|
| Concept Identification    | 25%       | Identifies all key concepts from literature accurately and comprehensively  | Identifies most key concepts with minor omissions | Identifies limited concepts; some are irrelevant or missing | Fails to identify essential concepts    |
| Linkages                  | 25%       | Establishes clear, meaningful, and accurate relationships between concepts  | Shows correct relationships with minor gaps       | Relationships are weak, unclear, or partially incorrect     | No meaningful relationships established |
| Organization              | 20%       | Concepts are well-structured hierarchically with logical flow and grouping  | Mostly organized with minor structural issues     | Some organization but lacks clear hierarchy                 | No logical organization or structure    |
| Integration of Literature | 20%       | Integrates multiple studies effectively to show connections and comparisons | Shows some integration of literature              | Limited integration; mostly isolated concepts               | No integration of literature evidence   |
| Clarity                   | 10%       | Concept map is clear, neat, visually structured, and easy to interpret      | Generally clear with minor readability issues     | Clarity is reduced; difficult to interpret in parts         | Poorly presented and hard to understand |

## 1.CONCEPT MAP



**Answer:**

- Missing concepts: Design and Deployment

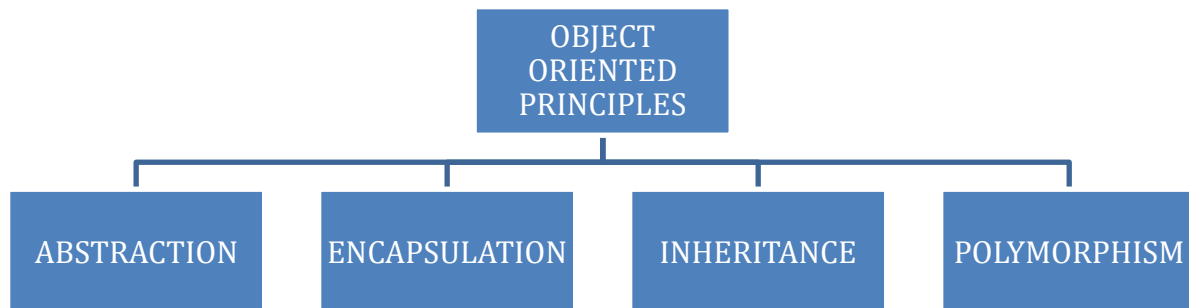
**Explanation:**

- Design converts requirements into the system architecture.
- Deployment releases the software to end users.

**Analysis:**

- Ensures systematic software development.
- Reduces project risks.
- Improves software quality and maintainability.

## 2. CONCEPT MAP



**Answer:**

- Missing concepts: Encapsulation and Polymorphism

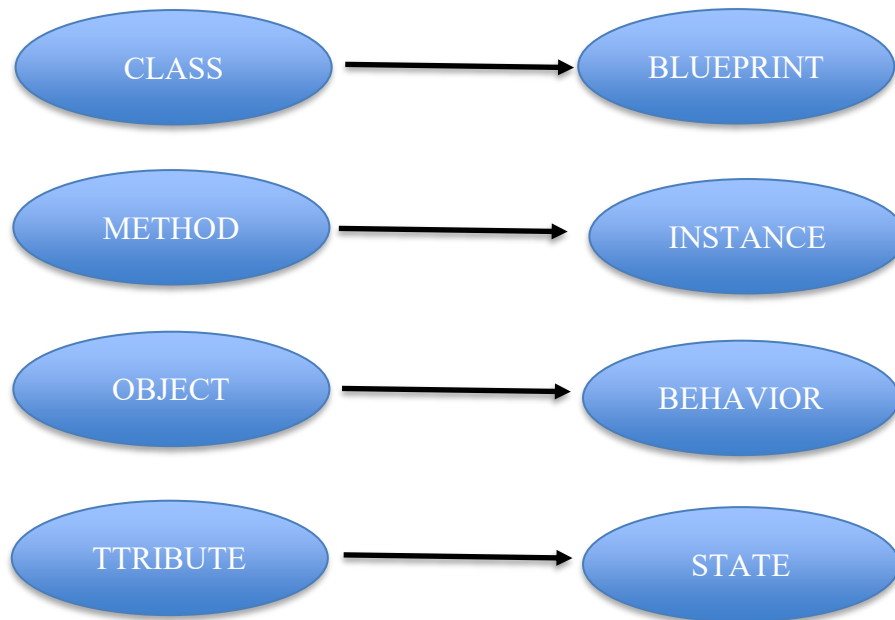
**Explanation:**

- Encapsulation protects data from unauthorized access.
- Polymorphism allows one interface to support multiple behaviors.

**Analysis:**

- Improves security and flexibility.
- Enhances code reusability.
- Supports efficient software design.

### 3. CONCEPT MAP



**Answer:**

- Missing concepts: Behavior and State

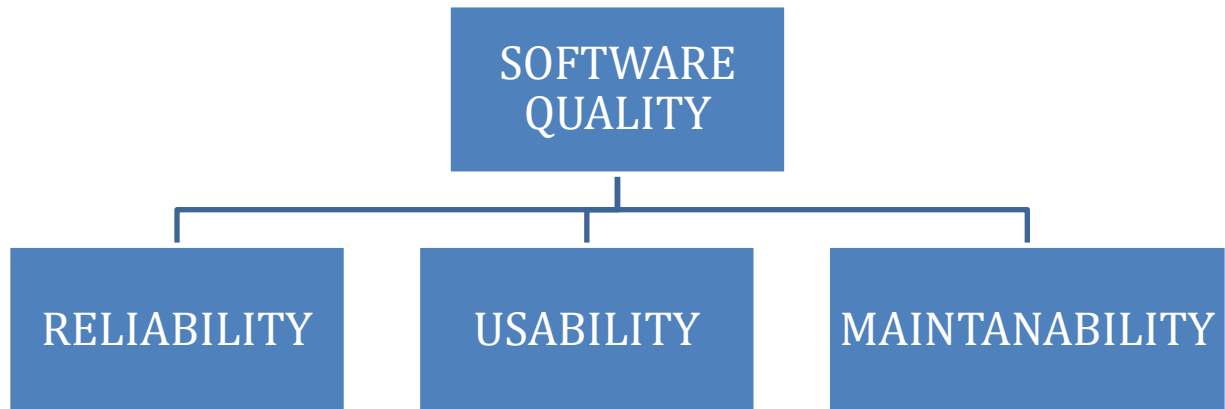
**Explanation:**

- Methods define the behavior of objects.
- Attributes represent the state of objects.

**Analysis:**

- Helps model real-world entities.
- Defines object characteristics and actions.
- Improves system organization.

### 4. CONCEPT MAP



**Answer:**

- Missing concept: Usability

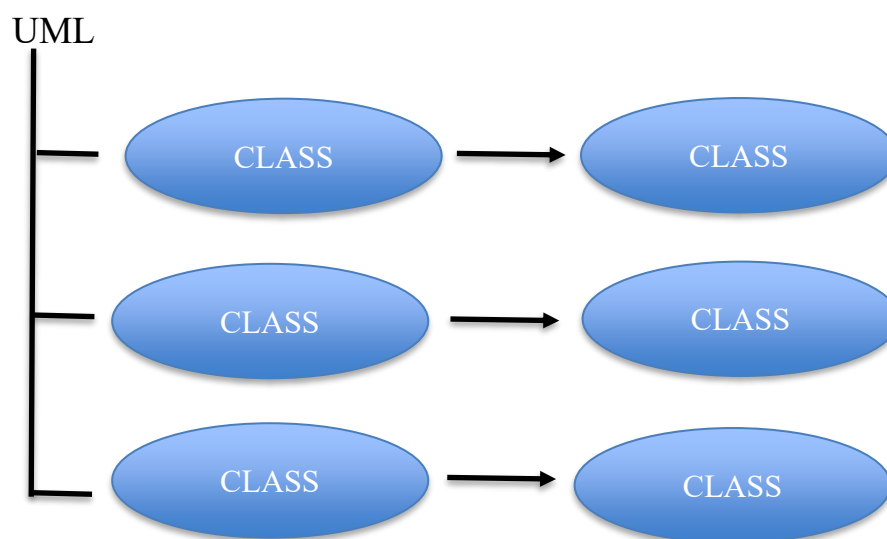
**Explanation:**

- Usability measures how easily users interact with software.
- Reliable software performs consistently.
- Maintainable software is easy to update.

**Analysis:**

- Improves user satisfaction.
- Enhances product acceptance.
- Contributes to software success.

## 5. CONCEPT MAP



**Answer:**

- Missing concepts: Object Diagram and Communication Diagram

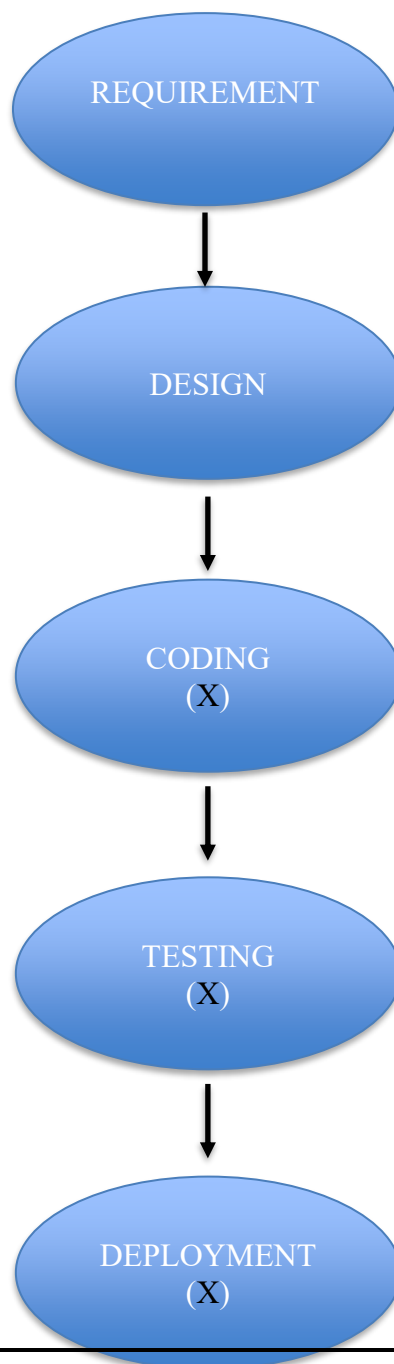
**Explanation:**

- Object diagrams represent object instances and relationships.
- Communication diagrams show message exchange among objects.

**Analysis:**

- Improves system visualization.
- Supports object-oriented analysis.
- Facilitates stakeholder communication.

## 6. CONCEPT MAP



**Explanation**

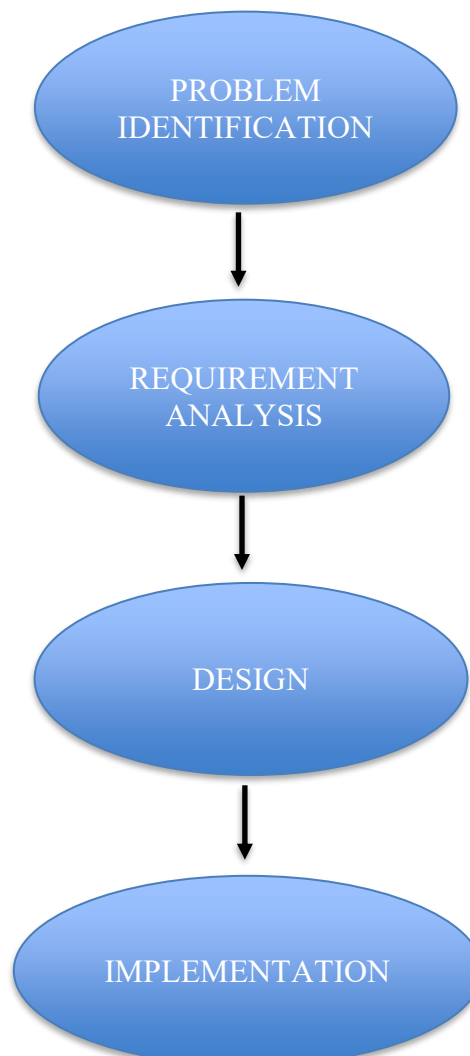
- Failure occurs during the Coding phase.
- Program logic contains implementation errors.

**Analysis**

- Errors affect subsequent phases.
- Testing cannot proceed effectively.
- Software quality decreases.

**Solution**

- Review source code.
- Apply coding standards.
- Conduct code inspections before testing.

**7. CONCEPT MAP**



**Answer**

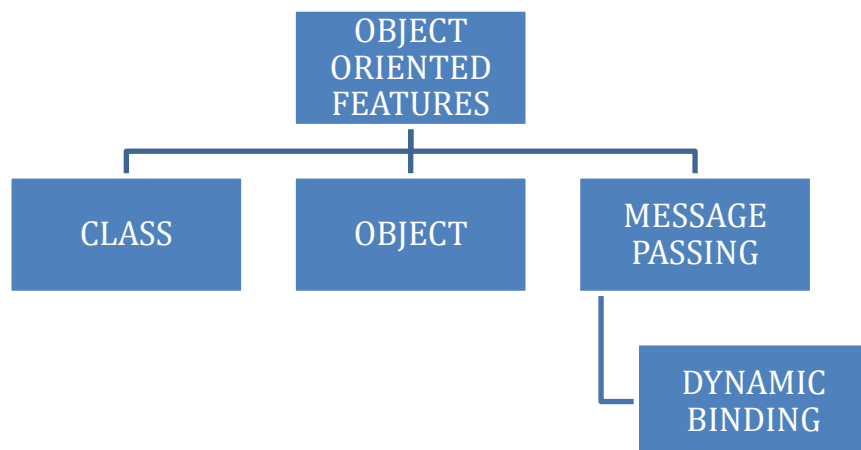
- Requirement Analysis
- Testing

**Explanation**

- Requirement Analysis identifies user needs and system requirements.
- Testing verifies whether the implemented system meets the specified requirements.

**Analysis**

- Improves software quality.
- Reduces development errors.
- Ensures successful project completion.

**8. CONCEPT MAP****Answer**

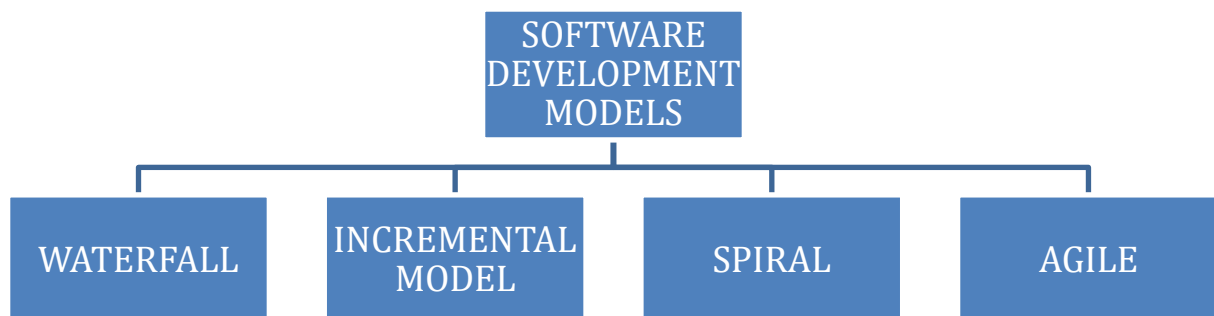
- Object
- Dynamic Binding

**Explanation**

- Objects are instances of classes.
- Dynamic binding allows method calls to be resolved at runtime.

**Analysis**

- Supports flexibility in software design.
- Improves code maintainability.
- Enables runtime polymorphism.

**9. CONCEPT MAP****Answer**

- Incremental Model
- Agile Model

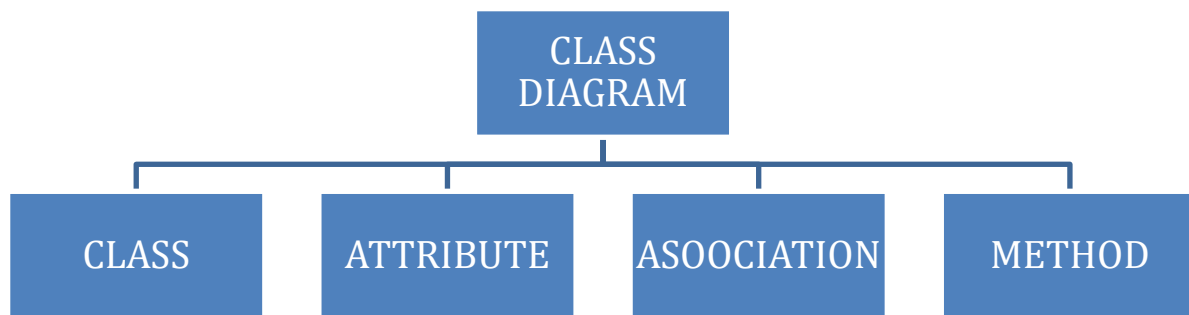
**Explanation**

- Incremental development delivers software in stages.
- Agile supports iterative development with continuous customer feedback.

**Analysis**

- Enhances adaptability.
- Reduces project risks.
- Improves customer satisfaction.

**10. CONCEPT MAP**



**Answer**

- Attribute
- Method

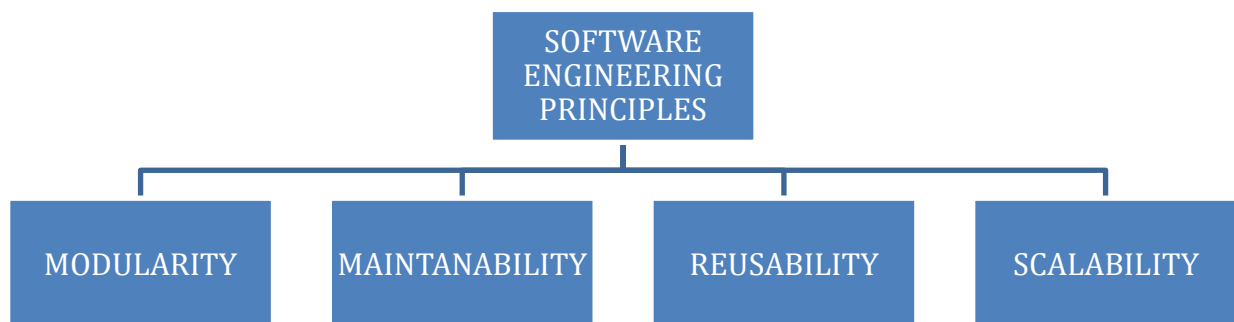
**Explanation**

- Attributes define the properties of a class.
- Methods define the operations performed by a class.

**Analysis**

- Represents object structure effectively.
- Improves software documentation.
- Supports object-oriented design.

**11. CONCEPT MAP**



**Answer**

- Maintainability
- Scalability

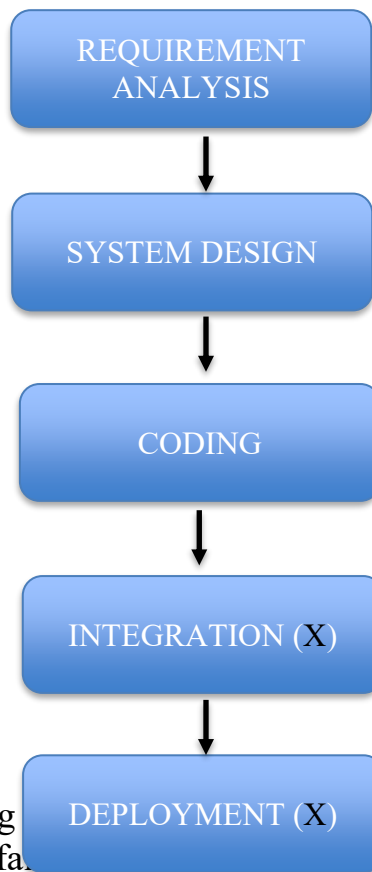
**Explanation**

- Maintainability simplifies software updates.
- Scalability enables systems to handle increased workload efficiently.

**Analysis**

- Improves software performance.
- Reduces maintenance effort.
- Supports long-term software evolution.

## 12. CONCEPT MAP



### Explanation

- Failure occurs during
- Individual modules fail to integrate correctly.

### Analysis

- Causes functional inconsistencies.
- Delays software deployment.
- Increases debugging effort.

### Solution

- Perform integration testing.
- Verify module interfaces.
- Resolve compatibility issues before deployment.